

भारतीय मानक
Indian Standard

IS 8682 : 2026

**डेयरी उद्योग के अपशिष्ट जल के
उपचार और निपटान — दिशानिर्देश**
(पहला पुनरीक्षण)

**Treatment and Disposal of
Effluents of Dairy Industry —
Guidelines**

(*First Revision*)

ICS 13.030.20

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Price Group 8

Water Pollution Control Sectional Committee, EED 03

FOREWORD

This Indian Standard (First Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Water Pollution Control Sectional Committee had been approved by the Environment and Ecology Department.

With rapid industrialization and urbanisation taking place all over the country, the number of dairies industries is sharply rising. The dairy industries uses large quantities of water for various purposes, resulting in the corresponding discharge of effluents. The standard is published with an objective to provide guidelines for treatment and disposal of dairy industry effluents. The methods have been selected keeping in view the practicability of their adoption in India. The available literature on the subject has been carefully reviewed for recommending appropriate treatment and disposal methods.

This standard was first published in 1977. Considering the technological advancements and insights gained during the last 5 decades, the Committee has decided to revise this standard.

In this revision, the following modifications have been incorporated:

- a) In clause [4](#), the figure of flow processes in dairy manufacturing plant has been added;
- b) In clause [4](#), additional utility systems that produces dairy effluent has also been added;
- c) Sources of effluents in dairy industry in [Table 2](#) has been updated;
- d) In clause [5.2](#), a note has been added on the variation in the quantities of wastewater generated from the dairy industry; and
- e) This standard has been revised incorporating clauses on waste reduction, by-product recovery in dairy industry along with relevant technical changes in throughout the document.

The composition of Committee responsible for revision of this standard is given in [Annex C](#).

Indian Standard

TREATMENT AND DISPOSAL OF EFFLUENTS OF DAIRY INDUSTRY — GUIDELINES

(First Revision)

1 SCOPE

This standard provides guideline for treatment of effluents from dairy industry.

2 REFERENCES

The standards listed in [Annex A](#) contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. Since all standards are subject to revision, users of this standard are encouraged to ascertain the applicability of the most recent edition of these standards.

3 TERMINOLOGY

For the purposes of this document, the definitions given in IS 7022 (Part 1), IS 8413 (Part 1), IS 8413 (Part 2) along with the following definitions, shall apply:

3.1 Oxidation Ditch — A continuous-loop channel with extended aeration.

3.2 Rotating Biological Contactors (RBC) — Discs with biofilm rotate through wastewater and air, facilitating aerobic treatment.

3.3 Sequencing Batch Reactor (SBR) — Fill and draw batch process where all treatment steps occur in a single reactor tank.

3.4 Facultative Ponds — Natural treatment in large, shallow basins under aerobic, anaerobic or facultative conditions.

3.5 Facultative Lagoon — Mechanically aerated ponds where aerobic microorganisms degrade organic matter.

3.6 Anaerobic Lagoon — Deep ponds where anaerobic bacteria decompose organic matter without oxygen.

3.7 Upflow Anaerobic Sludge Blanket (UASB) — Wastewater flows upward through anaerobic sludge bed, generating methane.

3.8 Moving Bed Biofilm Reactor (MBBR) — A hybrid system where free-floating biofilm carriers are suspended in an aeration tank, enhancing microbial growth on surfaces while maintaining mixed reactor conditions.

3.9 Membrane Bioreactor (MBR) — Combines activated sludge treatment with membrane filtration (micro/ultrafiltration) for solid-liquid separation, eliminating need for secondary clarifiers.

4 DAIRY PRODUCTS AND THEIR CHARACTERISTICS

4.1 The standards for various dairy products and their respective characteristics are given in [Annex B](#) for information.

4.2 Outline of processes that are involved in manufacture of dairy products is depicted in [Fig 1](#).

Some of the processes where effluents are generated is given below:

- a) Cleaning in place (CIP) system;
- b) Boiler house;
- c) Cooling water; and
- d) Water-softening plant.

5 SOURCES, QUANTITY AND CHARACTERISTICS OF EFFLUENTS

Effluents arising from dairy plants include water from cleaning of milk mays, equipment, bottles and floors. They may also contain small amounts of spilled milk, spoiled or sour milk, skimmed milk, whey and buttermilk. These effluents arise from:

- a) rinsing and washing of mays, bottles, tanks, drums, equipment, pipelines and floors;
- b) overflows, spillages and leakages from pumps and other equipment and; and
- c) entrainment during evaporation

5.1 Sources

Sources of wastewater are given process-wise in the following [Table 1](#).

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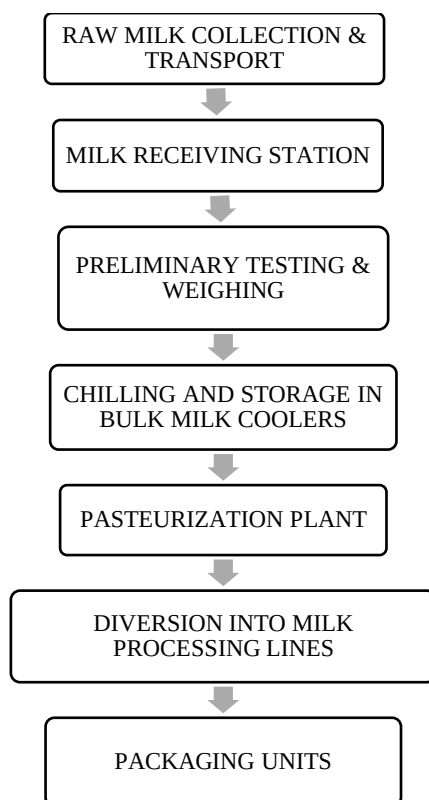


FIG. 1 FLOW OF PROCESSES INVOLVED IN A DAIRY MANUFACTURING PLANT

Table 1 Sources of Effluents in a Dairy

(Clause 5.1)

Sl No.	Source	Type and characteristics of effluent
(1)	(2)	(3)
i)	Milk receiving section	Milk drippings, rinses and washings from may washers
ii)	Pasteurization plant	Acidic/alkaline/detergent washings; floor washings; spills and leaks
iii)	Cooling water	Potential contamination due to leaks
iv)	Manufacturing of butter, buttermilk and ghee	Buttermilk, wash water from churns, traces of milk, skim milk and cream
v)	Manufacturing of cheese and curd	Whey, washings from vats, floors, presses; discarded whey and cleaning water
vi)	Manufacturing of flavoured milk and ice cream	Organic matter and suspended solids from milk and fruits (rich in fat and protein)
vii)	Protein powder and milk powder plant	Whey, mineral acids, food colouring agents and washings; batchwise and intermittent discharges
viii)	Clean-in-place (CIP) systems	Chemical washings from cleaning of tanks, pipes and equipment using alkalis, acids and sanitizers
ix)	Bottle-washing plant	Alkaline effluents due to detergents, caustic/washing soda; batchwise discharge (typically twice daily)
x)	Water-softening plant and boiler house	Mildly to highly acidic or saline effluents; boiler blowdown with suspended mineral solids

5.2 Quantity

On average, 3 litres of wastewater are generated per litre of processed milk, though this may range from 0.2 litres to 10 litres*.

5.3 Characteristics

5.3.1 Individual Effluents

The nature and composition of effluents from a dairy depend upon the quantity of milk processed and the type of products made.

5.3.1.1 The characteristics of individual effluents from a medium sized dairy processing 60 000 litres of milk per day (mostly bottling, preparation of butter and casein) are given in [Table 2](#).

5.3.1.2 Characteristics of individual effluents from a large dairy processing approximately 800 000 liters of milk per day and producing casein, butter, butteroil, ghee and, cheese are given in [Table 3](#).

5.3.2 Combined Dairy Effluent

The composition of the combined effluent from a dairy depends upon the quantity of milk processed and the types of products made. The characteristics of effluents from three different dairies are shown in [Table 4](#).

5.3.3 The biochemical oxygen demand (BOD) in pasteurization and bottling plants may be caused due to the loss of milk into the waste streams. Suspended solids may constitute a significant portion of the total solids in the combined effluent, while oil and grease levels are relatively high. Volatile solids account for about 70 percent to 90 percent of the total solids. The COD to BOD ratio of dairy effluents ranges from 1.4 to 1.6, indicating easy biodegradability. The BOD load may approximately be 2 kg to 8 kg per cubic meters of milk processed.

6 WASTE REDUCTION AND BYPRODUCT RECOVERY IN DAIRIES

6.1 Waste Reduction

Waste reduction in dairies may be effectively achieved by adopting the following measures:

- a) Allowing sufficient time for draining milk may, either manually or mechanically, to minimize residual losses;

- b) Using a small quantity of water in the form of a fine spray after initial draining, followed by additional draining time to remove remaining milk residues;
- c) Preventing the overflow of milk or other dairy products from vats, tanks and coolers through appropriate level control mechanisms;
- d) Preventing boil-over in vacuum pans by maintaining optimal temperature and pressure controls;
- e) Regularly inspecting and maintaining joints, packing glands and seals to prevent leakage of milk and other liquids;
- f) Preventing foaming in pipelines by eliminating air leaks in suction lines and minimizing splashing in tanks;
- g) Avoiding the entrainment of milk solids in condenser water from vacuum pans, as such losses may be as high as 1 percent of the total milk processed;
- h) Reducing the volume of wastewater by segregating clean water streams, such as condensates and cooling water, from strong waste streams;
- j) Implementing dry cleaning practices (for example, scraping and wiping) before wet washing to reduce water usage and load on effluent systems;
- k) Training staff regularly on best hygiene and operational practices to reduce spillage and wastage during processing;
- m) Automating valves, flow controls and CIP (clean-in-place) systems to minimize manual errors and ensure efficient cleaning with minimal waste; and
- n) Reusing process water where appropriate (for example, final rinse water for initial cleaning) in accordance with food safety regulation.

6.2 Reclamation of Waste Products

Wastewater and by-products from the dairy industry may be reclaimed and reused in various sustainable ways. The following practices may be adopted:

- a) Milk drippings and primary rinsing water (containing residual milk) may be collected and used as supplementary animal feed, especially for pigs and cattle;

* The wide range in dairy wastewater generation may be attributed to variations in product type and processing intensity — fluid milk plants may generate far less effluent than those producing cheese, butter or milk powder. Cleaning practices, particularly manual cleaning versus automated systems (such as CIP), may significantly influence water consumption. Larger, automated dairies with water reuse systems may discharge lower volumes of effluent, while small and medium-sized dairies may lack such infrastructure. Limited recycling and poor water management practices may further contribute to increased wastewater volumes. Additionally, human factors such as excessive cleaning, product spillage and inadequate staff training may also lead to higher effluent generation. Even within a single dairy, the effluent discharge rate may fluctuate significantly throughout the day.

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- b) Combined effluents (after preliminary or partial treatment) may be utilized in aquaculture, particularly for breeding carps and other tolerant fish species;
- c) Whey, after concentration and evaporation, may be processed into high-protein liquid or powder feed for livestock and poultry;
- d) Yeasts and other microbial biomass cultivated on whey substrates may be harvested, dried and used as a protein-rich animal feed supplement;
- e) Whey permeates (the lactose-rich liquid left after protein extraction) could serve as a fermentation substrate for producing bioethanol, lactic acid or bioplastics;
- f) Treated dairy effluent may be reused for irrigating non-edible crops (for example, fodder grass), subject to compliance with local effluent discharge and reuse standards; and
- g) Dairy sludge and high-strength effluents may be subjected to anaerobic digestion for biogas and manure production, supporting circular economy initiatives.

Table 2 Characteristics of Individual Effluents from a Medium Dairy

(Clause [5.3.1.1](#))

Sl No.	Characteristic	Butter Washings	Rejected Milk	Milk Bottling Effluent
(1)	(2)	(3)	(4)	(5)
i)	pH Value	4 to 7	6.6	7 to 8
ii)	Alkalinity as CaCO ₃ to pH 4.5 (mg/l)	100	10 350	256
iii)	Total solids (mg/l)	13 820	85 920	2 560
iv)	Suspended solids (mg/l)	5 112	34 904	1 380
v)	Biochemical oxygen demand for 5 days @ 20 °C (mg/l)	10 000	62 000	620
vi)	Chemical oxygen demand, (mg/l)	16 000	88 000	1 000

Table 3 Characteristics of Individual Effluents from a Large Dairy

(Clause [5.3.1.2](#))

Sl No.	Item	Cheese Plant	Milk Receiving and Pasteurization Units	Casein Plant	Butter, Butter oil and Ghee Section	Pooled Dairy Effluent
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	pH Value	7.2	8.1	7.6	7.9	6
ii)	Total solids	2 220	3 320	550	3 200	1 450
iii)	Suspended solids (mg/l)	598	1 270	105	2 140	585
iv)	Hardness (mg/l)	478	498	455	410	525
v)	Biochemical oxygen demand at 5 days and 20 °C(mg/l)	2 170	1 590	230	1 170	790
vi)	Chemical oxygen demand (mg/l)	3 090	2 540	350	3 170	1275
vii)	Oil and grease (mg/l)	534	657	540	1 150	310

Table 4 Characteristics of Composite Effluents from Three Dairies

(Clause 5.3.2)

SI No.	Item	Dairies		
		No. 1	No. 2	No. 3
(1)	(2)	(3)	(4)	(5)
i)	Average flow (L/day)	800 000 to 900 000	1 600 000 to 1 800 000	6 00 000
ii)	pH	7.0 to 7.5	12	7.5
iii)	Suspended solids (mg/l)	700	600	693
iv)	Dissolved solids (mg/l)	1 500	1 200	3 738
v)	Biochemical oxygen demand at 5 days and 20°C (mg/l)	1 200	1 000	1 180
vi)	Chemical oxygen Demand (mg/l)	2 500	2 000	2 328
vii)	Oil and grease (mg/l)	40 to 50	350	293

6.3 By-product Recovery

The recovery of usable by-products from dairy processing not only enhances resource efficiency but also helps reduce the pollution load in effluents. The major recovery processes may include the following:

- Buttermilk solids may be recovered through roller drying after preliminary condensation. The resulting dried buttermilk powder may be sold directly or used as a protein source in animal feed and food formulations;
- Whey generated during cheese production may be processed into a dry powder using single- or two-stage evaporators followed by spray drying. The final product may be used in animal feed, infant food or bakery products;
- Lactose from whey may be extracted and purified through crystallization. Alternatively, lactose-rich whey may serve as a growth medium for cultivating yeast or lactic acid bacteria, which may then be harvested for use in animal feed or fermented product industries;
- Casein may be extracted from spoiled or sour milk by acid precipitation, followed by washing and drying. The recovered casein is commonly used in the manufacture of plastics, paints, adhesives, textiles and resins;
- Butter that has become rancid or sour, along with deteriorated cream, may be

repurposed as raw material in the production of industrial fats or vegetable oil products, thereby reducing waste and recovering valuable fat content;

- During ghee production, a brownish residue rich in fat and protein is formed, which may be utilized in the preparation of sweets, animal feed or cosmetic products. After drying, it may also be used as a fuel; and
- Solid residues such as milk scum or boiler sludge may be used as compost or incorporated as binding agents in low-grade construction materials, depending on local practices and feasibility.

7 METHODS OF TREATMENT

7.1 Primary Treatment Methods

Primary treatment in dairy effluent management prepares the effluent for biological treatment by stabilizing flow and removing large solids, oil, grease and floatable matter that may interfere with microbial processes. Some common primary treatment methods for dairy effluent are:

7.1.1 Screening

Screens may be useful at the inlet of any treatment system where visible solids such as milk caps, plastic and packaging waste are present. They could be used for all dairy plants to protect mechanical equipment from clogging and abrasion.

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7.1.2 Grease Traps

Grease traps might be used in dairies producing high-fat products like cream, ghee or butter. They may help reduce oil and grease loads before biological treatment or reuse in agriculture.

7.1.3 Dissolved Air Flotation (DAF)

DAF units could be well-suited for dairies dealing with high-strength effluents containing emulsified fats, proteins or suspended solids-such as those from cheese or paneer production. They may be useful before secondary treatment to reduce load.

7.1.4 Simple Settling

Simple settling might be considered where space is available and the effluent contains occasional grit or heavier particles, though it may not be effective for typical dairy effluents with low settleable content.

7.1.5 Chemical Coagulation

This method may be appropriate where partial

removal of fine suspended and colloidal matter is needed, particularly in dairies aiming to meet stricter discharge standards. It might be applied selectively due to its cost and sludge production.

7.1.6 Equalization and Pre-aeration

Equalization and pre-aeration could be helpful in plants with intermittent operations or variable flow/quality-such as shift-based dairies. Aeration may also be beneficial where long retention times could lead to odour or septic conditions.

7.2 Secondary Treatment Methods

Secondary treatment may be recommended to reduce organic and nutrient loads using biological processes, helping dairies move closer to regulatory compliance. [Table 5](#) outlines some of the secondary treatment methods that may be used. For further design guidelines of trickling filters, activated sludge process and waste stabilization ponds refer IS 8413 (Part 1), IS 8413 (Part 2) and IS 5611 respectively.

Table 5 Secondary Treatment Methods for Treating Primary Effluent

(Clause [7.2](#))

Sl No.	Treatment Method	Design Parameters	Suitability	Removal Efficiency
(1)	(2)	(3)	(4)	(5)
i)	Activated sludge process	a) OLR: 0.3 kg to 0.6 kg BOD/m ³ ·d b) HRT: 4 h to 8 h c) Requires primary sedimentation, seed sludge adaptation, sludge digestion	May be suitable for larger dairies due to its compact layout and efficient BOD removal, though it might be energy-intensive and could be sensitive to shock loads	BOD reduction may range from 85 % to 95 %; COD removal could be about 90 %; TN reduction might reach 60 %
ii)	Trickling filter	a) OLR: 0.08 kg to 1.2 kg BOD/m ³ ·day b) HLR: 1 m ³ /m ² ·day to 40 m ³ /m ² ·day c) HRT: 6 h to 10 days	Could work well across different plant sizes and operation may be relatively simple, but it might require effective pretreatment and could lead to odour issues	BOD reduction may be up to 95 %
iii)	Oxidation ditch	a) OLR: 0.16 kg to 0.40 kg BOD/kg m ³ ·d b) HRT: 18 h to 24 h	Might offer reliable performance with low sludge generation, though it may demand considerable energy	BOD removal may fall between 95 % to 98 %
iv)	Rotating biological contactors (RBC)	a) OLR: 0.3 kg to 1.3 kg BOD/m ³ ·day b) HLR: 0.04 m ³ /m ² ·day to 2 m ³ /m ² ·day c) HRT: 8 h to 36 h d) Discs are rotated at 1 rpm to 2 rpm, with ~ 40 % of their diameter submerged, allowing biofilm alternately to contact wastewater and air	May operate with low energy input and simple mechanics, but installation could be capital-intensive and performance might depend on climatic conditions	COD reduction could be up to 90 %
v)	Sequencing batch reactor (SBR)	a) OLR: 1.3 kg BOD/m ³ ·day b) HRT: 6 h to 10 h per cycle	Could be compact and handle variable loads effectively, though it may involve more technical complexity compared to conventional systems	COD removal may vary from 90 % to 99 %; TKN reduction may be around 80 %

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Table 5 (Continued)

Sl No.	Treatment Method	Design Parameters	Suitability	Removal Efficiency
(1)	(2)	(3)	(4)	(5)
vi)	Facultative Ponds	a) OLR: 15 kg to 80 kg BOD/ha·day b) HLR: 0.1 m ³ /m ² ·day to 0.3 m ³ /m ² ·day c) HRT: 7 days to 30 days d) Area: 1 ha to 4 ha	Might be ideal for rural or low-resource settings due to their simplicity and low cost, although they could require large land areas and consistent sunlight	BOD reduction might range between 80 % to 95 %
vii)	Facultative Lagoon	a) OLR: 50 kg to 200 kg BOD/ha·day b) HLR: 0.2 to 0.5 m ³ /m ² ·day c) HRT: 7 days to 20 days d) Area: 1 to 4 ha	May suit moderately priced installations where aeration and settling are feasible, though performance could vary with environmental conditions	BOD removal may fall within 80 % to 95 %
viii)	Anaerobic Lagoon	a) OLR: 0.2 kg to 0.3 kg BOD/m ³ ·day b) HLR: 0.05 m ³ /m ² ·d to 0.15 m ³ /m ² ·day c) HRT: 4 days to 10 days	Might be a cost-effective option in warm climates, but could present issues related to odour and sludge management	BOD reduction may range from 75 % to 95 %
ix)	Upflow Anaerobic Sludge Blanket (UASB)	a) OLR: 2 kg to 10 kg COD/m ³ ·day b) HRT: 6 h to 10 h c) No HLR; vertical design	Could be a high-rate anaerobic option with potential for energy recovery, though it may be complex to manage and might emit odours	COD removal may be about 80 % to 90 %; BOD reduction may reach ~ 95%

Table 5 (Concluded)

Sl No.	Treatment Method	Design Parameters	Suitability	Removal Efficiency
(1)	(2)	(3)	(4)	(5)
x)	Moving bed biofilm reactor (MBBR)	a) OLR: 1 kg to 6 kg COD/m ³ ·day b) HRT: 4 h to 8 h c) Carrier fill: 30 % to 70 % of tank volume d) Aeration or mixing required to keep carriers in suspension	May be well-suited for limited-space or retrofit applications and could offer good resistance to load variations, though costs might be moderate	BOD reduction might range from 80 % to 95 %; COD removal could fall between 70 % to 90 %
xi)	Membrane bioreactor (MBR)	a) OLR: 0.5 kg to 2.0 kg BOD/m ³ ·day b) HRT: 4 h to 6 h c) Mixed liquor suspended solids: 8 000 mg/l to 12 000 mg/l d) Membrane pore size: 0.01 µm to 0.1 µm	Might be preferred for achieving high-quality effluent and for sites with limited land, though it could involve high energy use and maintenance requirements	BOD and COD reduction may exceed 95 %; TSS removal may be over 90 %; pathogen removal might go beyond 99 %

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7.3 Tertiary Treatment Methods

Tertiary (advanced) treatments may be used when effluent reuse, zero liquid discharge (ZLD) or the removal of specific pollutants (for example, nitrogen, phosphorus, TDS, pathogens, residual BOD/COD) is required or when high-quality discharge is needed for groundwater recharge, landscaping or to meet public health and aesthetic standards.

7.3.1 Electrocoagulation

Electrocoagulation uses electric current to destabilize and aggregate contaminants via in-situ coagulant generation from metal electrodes and removes dissolved organic waste, turbidity and colouring agents. It helps in the significant removal of suspended colloidal particles.

7.3.2 Adsorption

Activated carbon adsorption is used as a pretreatment step before membrane separation processes.

7.3.3 Membrane Treatment

Common membrane separation processes include microfiltration, nanofiltration, ultrafiltration, reverse osmosis and electrodialysis. These methods offer high product recovery and produce effluent of high quality suitable for direct reuse.

7.3.4 Filtration

Filtration, such as sand or activated carbon filters, removes turbidity, color and odor from the effluent. Sand filters physically trap suspended particles, while carbon filters adsorb dissolved organics. These methods are effective and easy to install, though media replacement and maintenance is needed.

7.3.5 Disinfection

Disinfection, using ultraviolet (UV) light or ozonation, may be applied as the final step to inactivate pathogens. UV may be chemical-free and fast, while ozonation might also improve color and odor. However, UV tends to be less effective with turbid water and ozone might require energy-

intensive generation.

7.3.6 Zero Liquid Discharge (ZLD)

ZLD is a high-end treatment approach where all wastewater is treated and recycled, leaving no liquid waste for discharge. It includes evaporation, crystallization and advanced membrane technologies like reverse osmosis. While environmentally sustainable, it is capital and energy-intensive and suitable mainly for large dairies with strict compliance or water scarcity issues.

8 METHODS OF DISPOSAL

8.1 Disposal onto Land

Land disposal dairy effluents may give good results provided planned irrigation system is designed properly and the effluent is sufficiently treated. Different irrigation systems that may be used are as follows:

- a) ridge and furrow;
- b) flood irrigation with under-drains;
- c) sprinkler irrigation; and
- d) drip irrigation.

Selection of an appropriate method of irrigation, based on soil type, land topography and crop water requirements, may be guided by IS 11711. To prevent clogging of emitters and nozzles in sprinkler and drip irrigation systems, the guidelines and specifications provided in IS 17644 may be referred to. The evaluation of the adopted irrigation system's effectiveness may be done on the basis of IS 13062.

8.2 Disposal into Inland Surface Water

Dairies should ensure that the wastewater, being discharged, is adequately treated so as to meet the standards as prescribed by regulatory authorities

8.3 Disposal into Sewer System

Effluent from dairy processing units may be discharged into town sewer systems leading to secondary treatment of stage as per the limits set by the concerned regulatory bodies.

ANNEX A

(Clause 2)

LIST OF REFERRED STANDARDS

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
IS 5611 : 1987	Code of practice for construction of waste stabilization ponds (facultative type)	IS 11077 : 1984	Glossary of terms on soil and water
IS 7022 (Part 1) : 2024	Water related — Glossary of terms: Part 1 Water, sewage and industrial effluents (<i>first revision</i>)	IS 11711 : 2025	Criteria for adoptability of different irrigation methods — Recommendation (<i>first revision</i>)
IS 7005 : 1973	Code for hygienic conditions for production, processing, transportation and distribution of milk	IS 13062 : 2026	Irrigation equipment and systems — Evaluation of field irrigation efficiencies — Guidelines (<i>first revision</i>)
IS 8413	Requirements for biological treatment and equipment :	IS 17664 : 2021	Treated wastewater reuse for irrigation — Guidelines for the adaptation of irrigation systems and practices to treated wastewater
(Part 1) : 1977	Trickling filter		
(Part 2) : 1982	Activated sludge process and its modifications		

To access Indian Standards click on the link below:
<https://standards.bis.gov.in/website/know-your-standards>

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ANNEX B

(Clause [4.1](#))

INDIAN STANDARDS FOR DAIRY PRODUCTS

B-1 The details of Indian Standards for dairy products is given below:

<i>Sl No.</i>	<i>Product/Processes</i>	<i>Ref to</i>
(1)	(2)	(3)
i)	Pasteurized milk (collection, homogenization and standardization)	IS 13688
ii)	Butter	IS 13690
iii)	Buttermilk	IS 18678
iv)	Ghee	IS 16326
v)	Curd	IS 9617
vi)	Condensed milk	IS 1166
vii)	Paneer	IS 10484
viii)	Cheese	IS 2785
ix)	Flavoured milk	IS 4709
x)	Infant food	IS 1656
xi)	Ice Cream and kulfi	IS 2802
xii)	Casein	IS 1167
xiii)	Yoghurt	IS 12898
xiv)	Shrikhand	IS 10974 (Part 5)
xv)	Khoa	IS 10947 (Part 1)
xvi)	Milk powders	IS 14542, IS 1165, IS 13334 (Part 1), IS 13334 (Part 2)

ANNEX C

(Foreword)

COMMITTEE COMPOSITION

Water Pollution Control Sectional Committee, EED 03

<i>Organization</i>	<i>Representative(s)</i>
CSIR - National Environmental Engineering Research Institute, Nagpur	DR S. VENKATA MOHAN (Chairperson)
Alkali Manufacturers Association of India, New Delhi	REPRESENTATIVE SHRI HARI SARAN DAS (<i>Alternate</i>)
Central Pollution Control Board, New Delhi	DR VISHAL GANDHI
Confederation of Indian Industry, New Delhi	SHRIMATI SHILPA NISCHAL SHRIMATI SIPIKA CHAUHAN (<i>Alternate</i>)
CSIR - Central Leather Research Institute, Chennai	DR K. SRI BALA KAMESWARI DR S. SWARNALATHA (<i>Alternate</i>)
CSIR - Central Salt and Marine Chemicals Research Institute, Bhavnagar	DR SOUMYA HALDAR DR HITESH THAKARSHIBHAI SARAVAIA (<i>Alternate</i>)
CSIR - National Environmental Engineering Research Institute, Nagpur	DR GIRISH R POPHALI DR RIMA BISWAS MONDAL (<i>Alternate</i>)
Engineers India Limited, Gurugram	SHRI CHIRANJIBI PATTANAIAK SHRI HIMANSHU GUPTA (<i>Alternate</i>)
Federation of Indian Chambers of Commerce and Industry, New Delhi	SHRI MRITYUNJAY KUMAR SHRIMATI TANYA KHANNA (<i>Alternate</i>)
Hindustan Unilever Limited, Mumbai	SHRI SANJAY HARLAKA
Indian Chemical Council, Mumbai	SHRI NEERAJ GARG SHRI DHRUMIL SONI (<i>Alternate</i>)
Indian Institute of Technology Bombay, Mumbai	PROF SUPARNA MUKHERJI DR RENUKA VERMA
Indian Institute of Technology Madras, Chennai	DR TANUSHREE PARSAI
Indian Oil Corporation Limited, New Delhi	SHRI RAVI PRAKASH GUPTA SHRI MANOJ KUMAR UPRETI (<i>Alternate</i>)
Ministry of Housing and Urban Affairs, New Delhi	SHRI ROHIT KAKKAR SHRI SATISH KUMAR (<i>Alternate</i>)
Ministry of Jal Shakti, New Delhi	SHRIMATI JYOTIKA PEGU

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